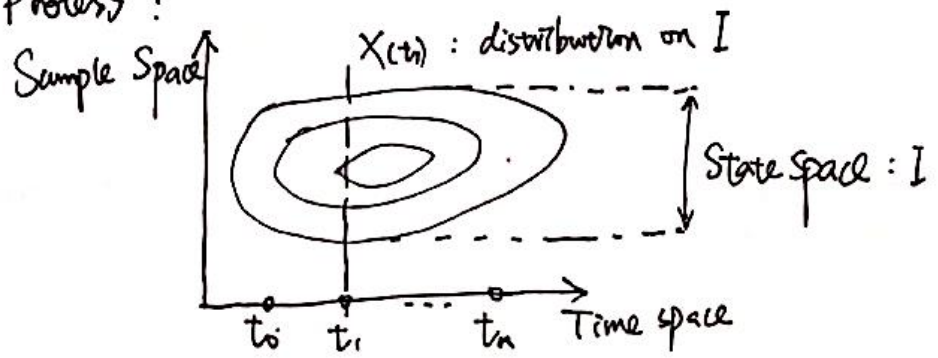
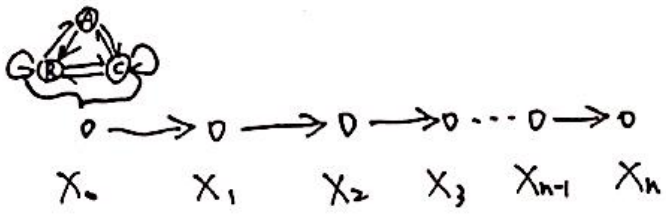
Random Process :



Function Families : $\{F_X(x;t) : t \in T\}$

Markov Chain :



(Burn-in + $\lambda A = \lambda$)

State space = $\{A, B, C\} = I$

Given – State of X_n = A or B or C

Given X_{n-1} $\begin{cases} \text{Markov: } P(X_n \mid X_0 X_1 \cdots X_{n-1}) = P(X_n \mid X_{n-1}) \\ P(X_{n-2} X_n \mid X_{n-1}) = P(X_{n-2} \mid X_{n-1}) \cdot P(X_n \mid X_{n-1}) \end{cases}$

State Transfer :

※ One-step : $P(X_{n+1} = A \mid X_n = B, X_{n-1} = C, X_{n-2} = A \cdots)$

$= P(X_{n+1} = A \mid X_n = B)$

$= P_{BA} (= M_{ST} \cdot [\begin{smallmatrix} 0 \\ 1 \\ 0 \end{smallmatrix}][0]) = M_{ST}(BA)$

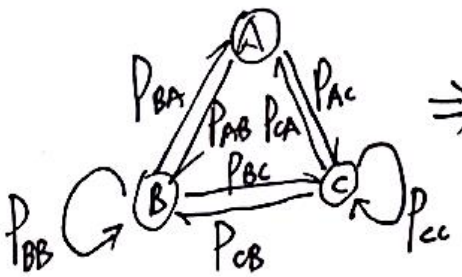
※ Multi-step : $P(X_{m+n} = A \mid X_m = B)$

$= \underbrace{M_{ST} \cdot M_{ST} \cdots M_{ST}}_{\times n} [\begin{smallmatrix} 0 \\ 1 \\ 0 \end{smallmatrix}][0]$

$= M_{ST}^{(n)}(BA)$

$= M_{ST}^{n}(BA)$ <or P_{BA}^{n}>

State Transfer Matrix : M_{ST}



$\Rightarrow$

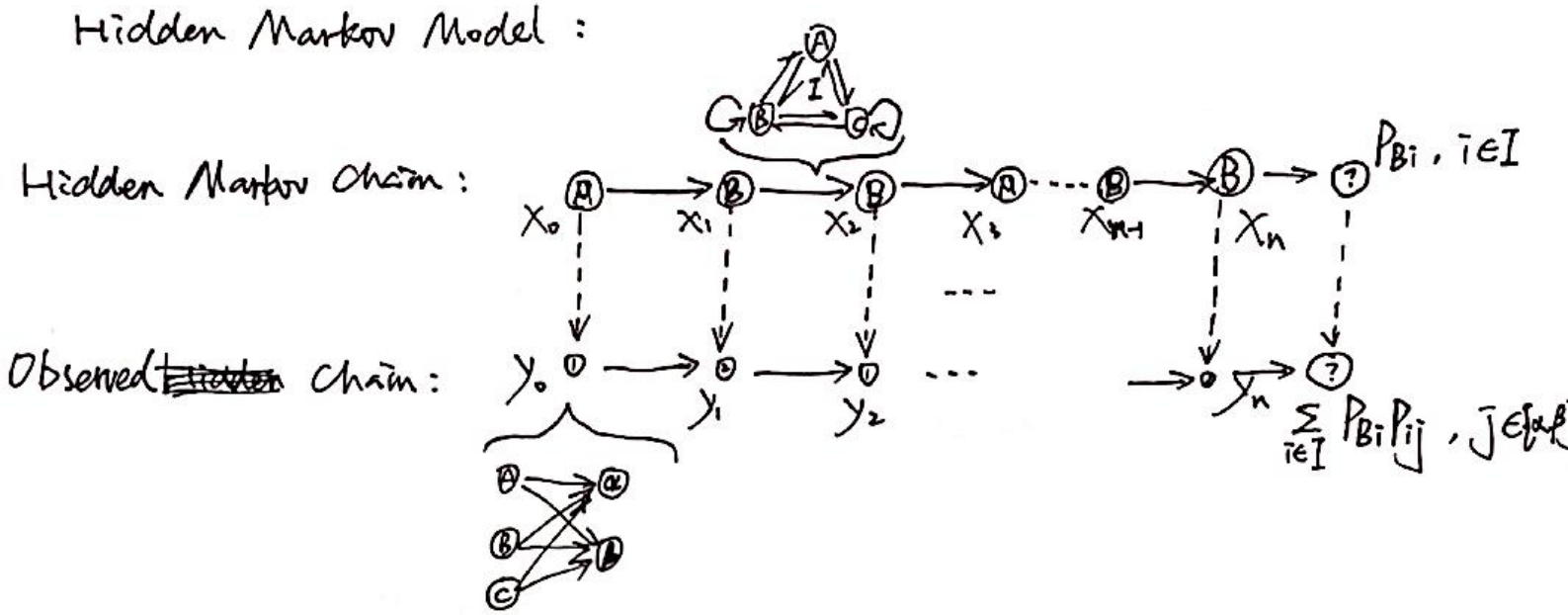
	A	B	C
A	P_{AA}	P_{BA}	P_{CA}
B	P_{AB}	P_{BB}	P_{CB}
C	P_{AC}	P_{BC}	P_{CC}

$P(X_n = A) = \sum_{i \in I} P(X_n = A \mid X_0 = i) P(X_0 = i)$

$= \sum_{i \in I} M_{ST}^{n}(iA) P(X_0 = i)$

Finite dimensional distribution

Hidden Markov Model :



Joint Probability :

$$P(X = A \to B \to C \ , \ y = \alpha \to \beta \to \alpha)$$

$$= \left[P(X_0 = A) \cdot P(y_0 = \alpha \mid X_0 = A)\right]\left[P(X_1 = B \mid X_0 = A)\, P(y_1 = \beta \mid X_1 = B)\right]\left[P(X_3 = C \mid X_2 = B) \cdots\right.$$

$$= P(X_0 = A) \cdot P_{known}$$

$\pi A = \pi$, Stable probability

Conditional Probability : (Posterior)

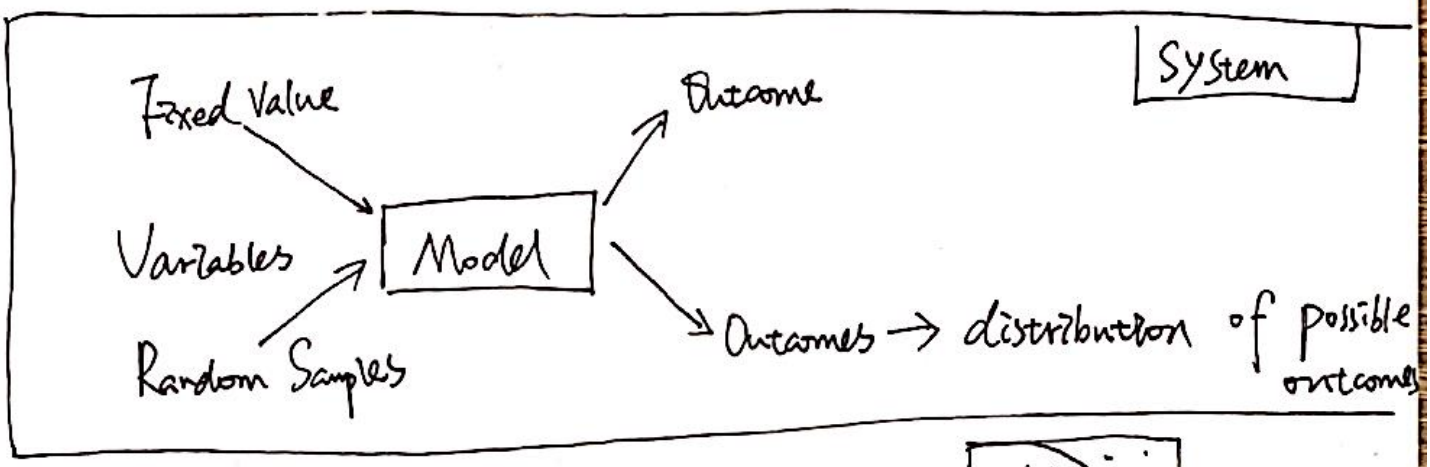
$$\cancel{\underset{x}{\operatorname{argmax}}}\ P(X \mid y = \alpha \to \beta \to \alpha)$$

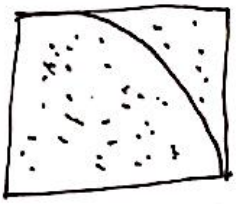
$$= \frac{P(y \mid x)\, P(x)}{P(y)} \propto P(y \mid x)\, P(x)$$

$$= \prod P(y_i \mid X_i) \cdot \prod P(X_i \mid X_{i-1})$$

$$\underset{X}{\operatorname{argmax}} \prod P(y_i \mid X_i) P(X_i \mid X_{i-1})$$ < Maximum Posterior >

Monte Carlo Method:



e.g. MKDA , π calculation  ...

Markov Monte-Carlo:

$$\underbrace{\circ \to \circ \to \circ \to \circ \to \cdots}_{\text{Burn in}} \to \underbrace{\circ \to \circ \to \circ \to \cdots}_{\text{Equilibrium distribution } P(x)}$$

$\pi A = \pi \quad (pT = p)$

Specific transition function $\longrightarrow$ desired target distribution

detailed balance: for targeted $P(x)$, transition $T(x|y)$ should:

$$P(x)\,T(y|x) = P(y)\,T(x|y) \quad , \forall x, y$$

$$\boxed{\begin{aligned} &\text{Metropolis: } T(x|y) = \min\left\{\frac{P(x)}{P(y)}, 1\right\} \\ &\text{Barker: } T(x|y) = \frac{P(x)}{P(x)+P(y)} \end{aligned}}$$

prove:

$$\sum_x P(x)\,T(y|x) = P(y)\sum_x T(x|y)$$
$$= P(y)$$
$$\Downarrow$$
$$pT = p$$